

THERE ARE NO RIESZ BASES OF EXPONENTIALS IN BALLS AND TRIANGLES

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ABSTRACT. We prove that balls in $\mathbb{R}^n$, $n \geq 2$, and nondegenerate planar triangles admit no Riesz basis of exponentials. The ball result extends to every bounded convex domain with C^2 boundary in dimensions at least two. The proof uses stationary limits and an obstruction for orthogonal projections. We also construct complete and minimal exponential systems with real frequencies on every bounded convex domain, with all biorthogonal functions supported in its John ellipsoid.

1. INTRODUCTION

Fix an integer $n \geq 2$. Let $\Omega \subset \mathbb{R}^n$ be an open ball and, for $\lambda \in \mathbb{R}^n$, write

$$e_\lambda(x) = e^{2\pi i \lambda \cdot x}.$$

We ask whether there is a discrete set $\Lambda \subset \mathbb{R}^n$ for which $\{e_\lambda\}_{\lambda \in \Lambda}$ is a Riesz basis of $L^2(\Omega)$. Thus the synthesis map

$$(c_\lambda) \mapsto \sum_{\lambda \in \Lambda} c_\lambda e_\lambda$$

is an isomorphism from $\ell^2(\Lambda)$ onto $L^2(\Omega)$; equivalently, the exponential system is complete and there are constants $0 < A \leq B < \infty$ such that

$$A \sum_{\lambda \in \Lambda} |c_\lambda|^2 \leq \left\| \sum_{\lambda \in \Lambda} c_\lambda e^{2\pi i \lambda \cdot x} \right\|_{L^2(\Omega)}^2 \leq B \sum_{\lambda \in \Lambda} |c_\lambda|^2 \quad (1.1)$$

for every finitely supported family (c_λ) .

The existence of an exponential Riesz basis for Euclidean balls is a longstanding folklore problem in harmonic analysis. The author learned about this problem from Yurii Lyubarskii in the late 90's. Bruna, Massaneda and Ortega-Cerdà [4, Section 5, p. 350] explicitly formulated

Date: September 25, 2026 (Version 2).

2020 Mathematics Subject Classification. 42C15, 42A55, 46E20.

Key words and phrases. Riesz basis, complete minimal exponential system, Paley–Wiener space, stationary projections, convex domain.

The author is supported by grant PID2024-160033NB-I00, funded by MICIU/AEI/10.13039/501100011033 and FEDER, UE, and by the Generalitat de Catalunya through grant 2024 ICREA 00142.

the conjecture that balls in dimensions greater than one have no such basis. The planar case was still recorded as open in 2025 [20, Section 1.2].

The main result is the following.

Theorem 1.1. *For every $n \geq 2$, no ball of positive radius in $\mathbb{R}^n$ admits a Riesz basis of exponentials.*

The same conclusion holds for every nonempty bounded convex open set with C^2 boundary in $\mathbb{R}^n$, $n \geq 2$; see Corollary 8.1 in Section 8. The argument also excludes exponential Riesz bases on every nondegenerate triangle in $\mathbb{R}^2$ (Theorem 7.1). More generally, Remark 6.4 states a boundary-measure criterion under which the same projection obstruction applies to other open sets.

The failure of exponential Riesz bases for balls, triangles and smooth convex domains makes it natural to ask whether these sets still admit complete and minimal exponential systems. Here minimality means that removing any one exponential destroys completeness, without a uniform stability requirement. Olevskii explicitly posed the general existence question in his 2021 Fields Institute lecture [23, Section 6, p. 7], for measurable sets $S \subset \mathbb{R}$ of finite positive measure. In Section 9 we prove the convex-domain analogue in every dimension: every nonempty bounded convex open set admits a complete and minimal system of exponentials with real frequencies (Theorem 9.1). No boundary regularity or symmetry is needed, and all the biorthogonal functions can be supported in the John ellipsoid of $\bar{\Omega}$. This result addresses the higher-dimensional convex case; the question for arbitrary measurable subsets of the line is not settled by this construction.

The orthogonal version of this problem was initiated by Fuglede [8]. Lev and Matolcsi [19] settled Fuglede's conjecture for convex domains in every dimension: a convex body admits an orthogonal basis of exponentials if and only if it tiles $\mathbb{R}^n$ by translations. Iosevich, Katz and Pedersen [12] proved that balls in dimensions at least two admit no orthogonal basis of exponentials. Quantitative restrictions on orthogonal exponential systems on a planar ball were obtained by Iosevich and Kolountzakis [13], and subsequently strengthened by Zakharov [28]. These results concern pairwise orthogonality. For Riesz bases, Kozma, Nitzan and Olevskii [17] constructed a bounded subset of the real line without an exponential Riesz basis and obtained a quantitative lower bound for the Riesz constant of a hypothetical basis on a planar ball, without excluding its existence.

Positive results are known for polygonal domains. Lyubarskii and Rashkovskii [21] constructed exponential Riesz bases for centrally symmetric convex polygons whose vertices lie on a lattice using entire functions of two complex variables. Debernardi and Lev [5] later removed the arithmetic restriction, proving existence for every centrally

symmetric convex polygon and, more generally, for centrally symmetric convex polytopes whose faces of all dimensions are centrally symmetric.

A related construction principle is lattice multi-tiling: translates of a bounded measurable domain by a lattice cover almost every point of $\mathbb{R}^n$ the same finite number of times. Grepstad and Lev [9] proved that such domains admit exponential Riesz bases under a Riemann measurability assumption; Kolountzakis [14] gave a simpler proof and removed this regularity assumption. Kozma and Nitzan proved existence for finite unions of intervals [15] and for finite unions of rectangles with sides parallel to the coordinate axes [16].

There is a function-theoretic reformulation of the problem. If

$$PW_\Omega = \left\{ F(z) = \int_\Omega f(x) e^{-2\pi i z \cdot x} dx : f \in L^2(\Omega) \right\}, \quad z \in \mathbb{C}^n,$$

with the norm transported from $L^2(\Omega)$, then PW_Ω is a Hilbert space of entire functions. For real λ ,

$$F(\lambda) = \langle f, e_\lambda \rangle_{L^2(\Omega)}.$$

Hence Λ is the frequency set of an exponential Riesz basis precisely when the restriction map

$$F \longmapsto (F(\lambda))_{\lambda \in \Lambda}$$

is an isomorphism from PW_Ω onto $\ell^2(\Lambda)$. Under the Fourier–Laplace identification, this restriction map is the adjoint of the synthesis operator. Thus the question is whether this Paley–Wiener space in n variables has a complete interpolating sequence.

The proof relies on a geometric feature of the ball: a nontrivial translate of its boundary meets the original boundary in a set of $(n-1)$ -dimensional surface measure zero.

The proof has four steps. First, separation of the frequency set allows us to replace its points by orthonormal translates of a small smooth bump. The Fourier transform of the bump is bounded away from zero on the domain, so the Riesz basis property yields an estimate $\|P - Q_\Lambda\| < 1$ between two orthogonal projections on $L^2(\mathbb{R}^n)$. Second, we take weak limits of translates of Λ and construct an invariant probability measure on the resulting compact space. This use of weak limits originates in Beurling’s work on sampling and interpolation [2]; a modern account of the topology is given in [10, Section 4]. Third, scalar Bochner measures and continuous box averages yield spectral cutoffs Π_t and comparison projections M_t on the associated stationary Hilbert space, with $\|\Pi_t - M_t\| \leq \gamma < 1$ for almost every t . We obtain this estimate by applying the original projection bound to compactly supported functions and averaging the resulting scalar inner products. Finally, crossing the boundary produces two strong limits of the cutoff projections with strictly nested ranges. Norm continuity of M_t places both limits at distance at most γ from M_{t_0} , giving a contradiction.

The translation-hull construction belongs to the established theory of point set dynamical systems; see Lenz–Stollmann [18] and Björklund, Hartnick and Pogorzelski [3]. Its applications to sampling and interpolation are developed by Enstad–Raum [6]. Averaging inequalities over invariant probability measures on hulls also plays a central role in [6, 7]. Here we use this framework to transfer a uniform distance estimate between projections to a stationary Hilbert space, where a boundary crossing produces the obstruction.

Sections 2–5 establish the analytic construction for any bounded open domain with boundary of Lebesgue measure zero. Section 6 supplies the geometric step for balls, Section 7 treats triangles, and Section 8 extends the result to nonempty bounded convex open sets with C^2 boundary. Section 9 proves the existence of complete minimal exponential systems on all bounded convex domains by a separate Fourier-analytic construction.

Note. The non-existence of Riesz bases of exponentials in balls and triangles has been independently obtained with a different proof by Zhiqiang Wan in [26]. We became aware of his result in the final stage of preparing our manuscript. Since the methods are different and our result also covers the full class of nonempty bounded convex open sets with C^2 boundary, a class not treated in [26], we offer this proof here.

Acknowledgments. The author thanks Joaquim Bruna for extensive discussions on the manuscript.

Use of artificial intelligence. A large language model was used for mathematical discussions and assistance with the Lean formalization. The work began with an unsuccessful attempt by the author to prove nonexistence of an exponential Riesz basis on the disk, following a strategy inspired by [22]. During these discussions, the model corrected the failed approach, leading to a proof based on representation theory. The author subsequently simplified that proof into the present form.

The Lean formalization of the nonexistence results in Sections 2–8 and the existence theorem in Section 9 is available at the GitHub repository: `RieszEuclidean`. The repository contains the proof blueprint, the Lean sources, a standalone proof file, and verification records identifying the statements checked by Comparator and Lean’s kernel. The existence theorem retains the Titchmarsh–Lions convolution support theorem as an explicit hypothesis. Its formal proof derives the required Bessel zero geometry directly from the radial differential equation. The verification records describe this alternative argument and its correspondence with the manuscript.

2. FROM COMPLETE INTERPOLATION TO BUMP PROJECTIONS

This section converts the Riesz basis assumption into a geometric statement about two closed subspaces of $L^2(\mathbb{R}^n)$. Disjoint smooth bumps model the coefficient space, and their projections onto the Fourier subspace reproduce the original synthesis operator up to an invertible multiplier.

Throughout Sections 2–5, Ω is a bounded open subset of $\mathbb{R}^n$ of positive measure. Inner products are linear in the first variable. We use the Fourier transform with the convention

$$(\mathcal{F}a)(x) = \int_{\mathbb{R}^n} a(\xi) e^{2\pi i \xi \cdot x} d\xi,$$

initially defined on $L^1 \cap L^2$, and set

$$P = \mathcal{F}^{-1} M_{1_\Omega} \mathcal{F}, \quad p(y) = \int_{\Omega} e^{-2\pi i y \cdot x} dx. \quad (2.1)$$

Thus P is an orthogonal projection on $L^2(\mathbb{R}^n)$ that commutes with translations. For compactly supported $a \in L^2(\mathbb{R}^n)$, Fourier inversion and Fubini's theorem give $(Pa)(v) = \int_{\mathbb{R}^n} p(v - w) a(w) dw$ almost everywhere.

2.1. Separated frequencies and an exact bump representation.

The lower Riesz bound first separates the frequencies. We then choose one sufficiently small bump so that its translates by the given frequencies are orthonormal and its Fourier transform has no zeros on $\overline{\Omega}$. This gives an equivalence between the Riesz basis property and a distance bound for projections, using the elementary criterion proved in Lemma 2.2 below.

Lemma 2.1. *The system $\{e_\lambda : \lambda \in \Lambda\}$ is a Riesz basis of $L^2(\Omega)$ if and only if there are $\delta, r > 0$, a real nonnegative $b \in C_c^\infty(\mathbb{R}^n)$ and $0 \leq \gamma < 1$ such that Λ is δ -separated,*

$$\|b\|_2 = 1, \quad \text{supp } b \subset B(0, r), \quad 2r < \delta, \quad \inf_{x \in \overline{\Omega}} |\mathcal{F}b(x)| > 0, \quad (2.2)$$

and the orthogonal projection Q_Λ onto the closed span of $\{b(\cdot - \lambda) : \lambda \in \Lambda\}$ satisfies

$$\|P - Q_\Lambda\| \leq \gamma < 1. \quad (2.3)$$

Proof. Suppose first that the system is a Riesz basis. Apply the lower Riesz bound to coefficients 1 and -1 at distinct frequencies. Since $|e^{iu} - e^{iv}| \leq |u - v|$,

$$2A \leq \|e_\lambda - e_\eta\|_{L^2(\Omega)}^2 \leq 4\pi^2 |\lambda - \eta|^2 \int_{\Omega} |x|^2 dx.$$

The integral is finite and positive. This gives a positive separation constant δ . Comparing the volumes of disjoint balls of radius $\delta/2$ also

gives a uniform bound on the number of frequencies in any ball of fixed radius.

Choose a real nonnegative smooth compactly supported b_0 with $\|b_0\|_2 = 1$. For small $\varepsilon > 0$ put $b(\xi) = \varepsilon^{-n/2} b_0(\xi/\varepsilon)$. Its support can be made arbitrarily small, and

$$(\mathcal{F}b)(x) = \varepsilon^{n/2} (\mathcal{F}b_0)(\varepsilon x).$$

Since $(\mathcal{F}b_0)(0) > 0$ and $\overline{\Omega}$ is compact, this proves (2.2) for a sufficiently small fixed ε .

For either direction, the conditions on b and Λ imply that the translates $b_\lambda = b(\cdot - \lambda)$ have disjoint supports and norm one. Consequently

$$W_\Lambda : \ell^2(\Lambda) \longrightarrow \text{Ran } Q_\Lambda, \quad W_\Lambda c = \sum_{\lambda \in \Lambda} c_\lambda b_\lambda,$$

is unitary. For finitely supported c ,

$$\mathcal{F}(PW_\Lambda c) = \mathbf{1}_\Omega(\mathcal{F}b) \sum_{\lambda \in \Lambda} c_\lambda e_\lambda. \quad (2.4)$$

Multiplication by $\mathcal{F}b$ is an isomorphism of $L^2(\Omega)$. If the exponential system is a Riesz basis, its synthesis operator is also an isomorphism onto $L^2(\Omega)$. Thus (2.4) extends to $\ell^2(\Lambda)$ and shows that $P : \text{Ran } Q_\Lambda \rightarrow \text{Ran } P$ is an isomorphism. Lemma 2.2 below gives (2.3).

Conversely, suppose the stated separation, bump and distance conditions hold. The same lemma shows that $P : \text{Ran } Q_\Lambda \rightarrow \text{Ran } P$ is an isomorphism. Hence

$$c \longmapsto (\mathcal{F}b)^{-1} \mathcal{F}(PW_\Lambda c)|_\Omega$$

is a bounded isomorphism from $\ell^2(\Lambda)$ onto $L^2(\Omega)$. By (2.4), it agrees with exponential synthesis on finitely supported sequences. The exponential system is therefore a Riesz basis. $\square$

2.2. Distance between orthogonal projections. We need two elementary facts about orthogonal projections. A distance strictly smaller than one is equivalent to invertibility of the projection between the ranges; consequently, two strictly nested ranges cannot both be this close to the range of a third projection.

For orthogonal projections P and Q on the same Hilbert space, we use the operator-norm distance $\|P - Q\|$.

Lemma 2.2. *Let P and Q be orthogonal projections on a Hilbert space. The map*

$$P : \text{Ran } Q \longrightarrow \text{Ran } P$$

is a bounded isomorphism if and only if

$$\|P - Q\| < 1. \quad (2.5)$$

Proof. Set $A = P|_{\text{Ran } Q} : \text{Ran } Q \rightarrow \text{Ran } P$. Its adjoint is $A^* = Q|_{\text{Ran } P}$. We first record the identity

$$\|P - Q\| = \max\{\|(I - P)Q\|, \|(I - Q)P\|\}.$$

Indeed, for every x ,

$$(P - Q)x = P(I - Q)x - (I - P)Qx.$$

The two terms on the right are orthogonal, as are $(I - Q)x$ and Qx . Since $\|P(I - Q)\| = \|(I - Q)P\|$, Pythagoras gives the upper bound in the identity. The reverse bound follows by multiplying $P - Q$ on the left by $I - P$ and $I - Q$.

Suppose first that $\|P - Q\| = \gamma < 1$. For $u \in \text{Ran } Q$ and $y \in \text{Ran } P$, Pythagoras gives

$$\|Au\|^2 \geq (1 - \gamma^2)\|u\|^2, \quad \|A^*y\|^2 \geq (1 - \gamma^2)\|y\|^2.$$

Thus A is injective with closed range, while $(\text{Ran } A)^\perp = \ker A^* = \{0\}$ makes its range dense in $\text{Ran } P$. Hence A is onto and has a bounded inverse.

Conversely, suppose A is an isomorphism. If both ranges are zero, the conclusion is immediate. Otherwise, put $c = \|A^{-1}\|^{-1} \in (0, 1]$. Since $(A^*)^{-1} = (A^{-1})^*$, both A and A^* are bounded below by c . Pythagoras therefore yields

$$\|(I - P)Q\| \leq \sqrt{1 - c^2}, \quad \|(I - Q)P\| \leq \sqrt{1 - c^2}.$$

The identity above gives $\|P - Q\| \leq \sqrt{1 - c^2} < 1$. □

The following consequence will give the final contradiction.

Lemma 2.3. *Let R_- and R_+ be orthogonal projections with*

$$\text{Ran } R_- \subsetneq \text{Ran } R_+.$$

There is no orthogonal projection M such that

$$\|R_- - M\| < 1, \quad \|R_+ - M\| < 1.$$

Proof. By Lemma 2.2, M maps each of the two ranges isomorphically onto $\text{Ran } M$. Choose $w \in \text{Ran } R_+ \setminus \text{Ran } R_-$. Surjectivity on the smaller range gives $u \in \text{Ran } R_-$ with $Mu = Mw$. Then $0 \neq w - u \in \text{Ran } R_+$ and $M(w - u) = 0$, contradicting injectivity on $\text{Ran } R_+$. □

Fix b, δ, r and γ from Lemma 2.1. For every δ -separated set Γ , let Q_Γ be the orthogonal projection onto the closed span of the corresponding bumps. Explicitly,

$$Q_\Gamma a = \sum_{\lambda \in \Gamma} \langle a, b_\lambda \rangle b_\lambda. \tag{2.6}$$

3. BEURLING WEAK LIMITS AND A STATIONARY SYSTEM

We enlarge the original frequency set to the compact space of all its translation limits. The projection estimate persists throughout this space. Averaging translates then provides an invariant probability measure and a Hilbert space with a nonzero translation-invariant vector.

3.1. Weak convergence of separated sets. Uniform separation prevents points from colliding in a limit. It also makes local convergence of configurations equivalent to vague convergence of their counting measures, which provides a convenient route to compactness.

We use Beurling's notion of weak convergence of point sets [2]. For a modern treatment see [10, Section 4]. A sequence of closed sets Γ_j converges weakly to Γ if, for every $R, \varepsilon > 0$, eventually

$$\Gamma_j \cap B(0, R) \subset \Gamma + B(0, \varepsilon), \quad \Gamma \cap B(0, R) \subset \Gamma_j + B(0, \varepsilon). \quad (3.1)$$

For sequences with a common positive separation constant, this is equivalent to vague convergence of their counting measures $\eta_\Gamma = \sum_{\lambda \in \Gamma} \delta_\lambda$. To see this, work in bounded regions whose boundaries contain no points of the limit set. Weak convergence matches the finitely many points in each such region, so continuous compactly supported test functions give vague convergence. Conversely, testing on small disjoint balls around limit points forces the matching points to exist. A sequence of unmatched points in a fixed compact set would have a convergent subsequence and would be detected by another nonnegative continuous test function. Uniform separation prevents more than one point from approaching a limit point.

The topology in (3.1) is the restriction of the Chabauty–Fell topology; see [3, Appendix A, Proposition A.1]. Lemma 3.1 is a standard consequence of compactness and metrizability of that space and continuity of its translation action: [6, Proposition 2.2(ii)], with $U = B(0, \delta/2)$, shows that the class of δ -separated sets is closed. See also [18, Theorem 1.2, Proposition 1.3 and Remark 1.5(c)] for the Euclidean topology and hull construction. We include a proof using counting measures.

Lemma 3.1. *The space of δ -separated subsets of $\mathbb{R}^n$, including the empty set, is compact and metrizable in the topology (3.1). The translation action is jointly continuous in the translation parameter and the configuration.*

Proof. The separation bound gives a uniform bound on the mass of the counting measures on each compact set. A diagonal extraction against a countable dense collection of continuous compactly supported functions gives a vaguely convergent subsequence, with a locally finite positive limit measure η .

We verify that η is again a counting measure. If $x \in \text{supp } \eta$, choose arbitrarily small balls $B(x, \varepsilon)$ of diameter less than δ whose boundaries

have η -measure zero. Vague convergence gives

$$\eta_{\Gamma_j}(B(x, \varepsilon)) \longrightarrow \eta(B(x, \varepsilon)).$$

The left-hand side takes values in $\{0, 1\}$, while the limit is positive because $x \in \text{supp } \eta$. Hence every such ball has mass one. Letting $\varepsilon \downarrow 0$ gives $\eta(\{x\}) = 1$. Thus η is the counting measure of its support. Two distinct support points at distance less than δ would, for large j , force two points of Γ_j to be less than δ apart. The support is therefore δ -separated; the zero measure corresponds to the empty set.

The same countable test functions metrize vague convergence on this uniformly locally bounded family. Sequential compactness therefore gives compactness. Finally, if $\Gamma_j \rightarrow \Gamma$ and $y_j \rightarrow y$, the local matching criterion gives $\Gamma_j - y_j \rightarrow \Gamma - y$. $\square$

Define the translation hull

$$X = \overline{\{\Lambda - y : y \in \mathbb{R}^n\}}, \quad T_y \Gamma = \Gamma - y.$$

This is a compact metrizable translation-invariant space. Its elements are precisely the weak limits of sequences of translates of Λ .

3.2. Passing the projection estimate to weak limits. Only finitely many bumps act on a compactly supported test function, so local convergence of the centers gives strong convergence of the bump projections. We also record a basic fact about strong limits that will avoid repeating the same argument in the later constructions.

A sequence of operators A_j converges *strongly* to A if $\|A_j f - A f\| \rightarrow 0$ for every fixed vector f . We use the same terminology for operator families depending on a real parameter. Strong convergence is preserved under products of uniformly bounded families: if $A_j \rightarrow A$ and $B_j \rightarrow B$ strongly, then

$$\|(A_j B_j - AB)f\| \leq \|A_j\| \|(B_j - B)f\| + \|(A_j - A)Bf\| \longrightarrow 0. \quad (3.2)$$

Self-adjointness passes to strong limits through inner products. Consequently, a strong limit of orthogonal projections is an orthogonal projection: the projections are uniformly bounded, and (3.2) preserves idempotence.

The local matching of centers used in the next proof is the separated case of [6, Proposition 2.3]; see also [7, Lemma 2.2]. Compact support of the bumps then gives strong convergence of the associated projections.

Lemma 3.2. *If $\Gamma_j \rightarrow \Gamma$ in X , then $Q_{\Gamma_j} \rightarrow Q_\Gamma$ strongly on $L^2(\mathbb{R}^n)$. In particular,*

$$\|P - Q_\Gamma\| \leq \gamma \quad (\Gamma \in X). \quad (3.3)$$

Proof. First take a compactly supported $a \in L^2(\mathbb{R}^n)$. Choose a ball containing the closed r -neighborhood of its support in its interior and having no point of Γ on its boundary. Within this ball, weak convergence and separation give, for large j , a bijection between the finitely many

centers of Γ_j and Γ , with matched centers converging. Only centers in this ball can contribute to (2.6). Translations of b are continuous in L^2 , so each term of this finite sum converges in L^2 . Thus $Q_{\Gamma_j}a \rightarrow Q_{\Gamma}a$. Compactly supported functions are dense, and all these projections have norm at most one, proving strong convergence.

Conjugating by translations preserves P and sends Q_{Λ} to the projection for the translated set. Thus every translate satisfies (2.3). Taking a strong limit and testing on an arbitrary vector gives (3.3). $\square$

Since P is nonzero, the empty set cannot belong to X : its projection is zero and would have distance one from P . By Lemma 2.2 and (2.4), each configuration in X is again the frequency set of an exponential Riesz basis for $L^2(\Omega)$. Preservation of the Riesz and frame properties under hull limits is also established in [6, Proposition 3.8 and Theorem 3.9]; for separated sets the bounds are preserved without loss. Only the uniform projection estimate will be needed below.

3.3. Translation averaging. Averaging the orbit over expanding boxes produces an invariant probability measure. The associated translation operators are strongly continuous, and the constant function has norm one. This invariant vector will detect the spectral jump at the boundary.

The following construction is the usual Følner averaging argument for an invariant probability measure on a compact hull. In the sampling setting, see [6, proof of Theorem 2.16, Steps 2–3]; see also the amenability discussion in [7, Section 2.3]. Neither unique ergodicity nor finite local complexity is needed here.

Put $C_R = [-R, R]^n$ and define a probability measure on X by

$$\mu_R = \frac{1}{|C_R|} \int_{C_R} \delta_{T_v \Lambda} dv.$$

Compactness gives a weakly convergent subsequence as $R \rightarrow \infty$; write μ for one such limit. No uniqueness is needed. For $F \in C(X)$ and fixed y ,

$$\left| \int F \circ T_y d\mu_R - \int F d\mu_R \right| \leq \|F\|_{\infty} \frac{|(C_R + y) \Delta C_R|}{|C_R|} \rightarrow 0.$$

Passing to the limit shows that μ is invariant under every T_y . Let

$$\mathcal{H} = L^2(X, \mu), \quad (U_y f)(\Gamma) = f(\Gamma + y). \quad (3.4)$$

Thus $U_y f = f \circ T_{-y}$, and the operators U_y form a unitary representation of $\mathbb{R}^n$. Joint continuity of the action on the compact space X makes $y \mapsto U_y f$ continuous in the uniform norm for $f \in C(X)$. Density of $C(X)$ in $\mathcal{H}$, together with unitarity, extends strong continuity to every $f \in \mathcal{H}$. The Hilbert space $\mathcal{H}$ is separable because X is compact and metrizable. The constant function $\mathbf{1}$ has norm one and is fixed by every U_y .

4. BOCHNER MEASURES AND STATIONARY FOURIER PROJECTIONS

We now construct the stationary analogue of the Fourier projection P . Scalar Bochner measures describe the frequencies of each vector, and Fejér averages approximate the indicator of Ω . A single control measure makes the exceptional set of translation parameters independent of the vector.

From now on, assume in addition that $|\partial\Omega| = 0$. We seek projections on $\mathcal{H}$ that retain the stationary frequencies θ for which $t + \theta \in \Omega$. Both the parameter t and the spectral variable θ belong to $\mathbb{R}^n$.

4.1. Scalar Fourier measures. Bochner's theorem associates a positive measure to the correlation function of each vector. The resulting norm identity turns convolution kernels into spectral multipliers; it is all the Fourier calculus needed below.

For $f \in \mathcal{H}$, the continuous function $y \mapsto \langle U_y f, f \rangle$ is positive definite. Bochner's theorem [25, Section 1.4.3, p. 19] gives a unique finite positive Borel measure σ_f on $\mathbb{R}^n$ such that

$$\langle U_y f, f \rangle = \int_{\mathbb{R}^n} e^{2\pi i y \cdot \theta} d\sigma_f(\theta), \quad \sigma_f(\mathbb{R}^n) = \|f\|^2. \quad (4.1)$$

Uniqueness and the parallelogram identity give

$$\sigma_{f+g} + \sigma_{f-g} = 2\sigma_f + 2\sigma_g, \quad \sigma_{cf} = |c|^2 \sigma_f. \quad (4.2)$$

Since $U_y \mathbf{1} = \mathbf{1}$, uniqueness also gives

$$\sigma_{\mathbf{1}} = \delta_0. \quad (4.3)$$

This atom at zero will ensure that the boundary crossing produces a nonzero change in the projection.

For $a \in L^1(\mathbb{R}^n)$ define

$$T_a f = \int_{\mathbb{R}^n} a(y) U_y f dy, \quad \phi_a(\theta) = \int_{\mathbb{R}^n} a(y) e^{2\pi i y \cdot \theta} dy.$$

We call a the *kernel* of T_a in translation coordinates and ϕ_a its *symbol* in frequency coordinates. The integral defining $T_a f$ is a Bochner integral in $\mathcal{H}$, with norm at most $\|a\|_1 \|f\|$. Expanding its squared norm and using (4.1) gives

$$\begin{aligned} \|T_a f\|^2 &= \iint a(y) \overline{a(w)} \langle U_{y-w} f, f \rangle dy dw \\ &= \int_{\mathbb{R}^n} |\phi_a(\theta)|^2 d\sigma_f(\theta). \end{aligned}$$

Fubini applies because $a \in L^1$ and σ_f is finite. The group law and unitarity give, for $a, c \in L^1(\mathbb{R}^n)$,

$$\|T_a f\|^2 = \int |\phi_a|^2 d\sigma_f, \quad T_a T_c = T_{a*c}, \quad T_a^* = T_{a^*}, \quad (4.4)$$

where $a^*(y) = \overline{a(-y)}$. In particular $\|T_a\| \leq \|\phi_a\|_\infty$. Convolution of kernels corresponds to multiplication of symbols, and $\phi_{a^*} = \overline{\phi_a}$. If two kernels have the same symbol, their operators agree by the norm identity. These facts provide the scalar calculus used below.

4.2. One measure to control all exceptional sets. Different vectors may have different spectral measures. A weighted sum over a countable dense set gives one measure whose null sets are null for every vector, allowing us to choose the later boundary point only once.

Choose a dense sequence $(h_j)_{j \geq 1}$ in the unit sphere of $\mathcal{H}$ and set

$$\sigma = \sum_{j=1}^{\infty} 2^{-j} \sigma_{h_j}. \quad (4.5)$$

This is a positive measure of mass one. For every Borel set E ,

$$\sigma(E) = 0 \implies \sigma_f(E) = 0 \quad (f \in \mathcal{H}). \quad (4.6)$$

Indeed, approximate f by scalar multiples g of the h_j . Then $\sigma_g(E) = 0$ and (4.2), applied to $f - g$ and g , gives

$$\sigma_f(E) \leq 2\sigma_{f-g}(E) + 2\sigma_g(E) \leq 2\|f - g\|^2 \longrightarrow 0.$$

Thus a single null-set condition applies simultaneously to all vectors in $\mathcal{H}$.

4.3. Continuous Fejér filters. The overlap of a box with its translates generates a nonnegative approximate identity. Its compactly supported inverse Fourier kernel serves two purposes: it defines bounded stationary multipliers here and reappears in the spatial averages of the next section.

The normalized overlap of the boxes $C_R = [-R, R]^n$ is

$$w_R(y) = \frac{|C_R \cap (C_R - y)|}{|C_R|} = \prod_{j=1}^n \left(1 - \frac{|y_j|}{2R}\right)_+. \quad (4.7)$$

Put

$$K_R(x) = \frac{1}{|C_R|} \left| \int_{C_R} e^{2\pi i v \cdot x} dv \right|^2, \quad m_R = K_R * \mathbf{1}_\Omega.$$

Plancherel gives $K_R \geq 0$ and $\int K_R = 1$. The scaling identity $K_R(x) = R^n K_1(Rx)$ shows that for each $\varepsilon > 0$, $\int_{|x| > \varepsilon} K_R(x) dx \rightarrow 0$. Thus $(K_R)_{R > 0}$ is an approximate identity as $R \rightarrow \infty$. Since $\mathbf{1}_\Omega$ is locally constant away from its boundary,

$$0 \leq m_R \leq 1, \quad m_R(x) \longrightarrow \mathbf{1}_\Omega(x) \quad (x \notin \partial\Omega). \quad (4.8)$$

Expanding the square in K_R shows that $K_R = \mathcal{F}w_R$. Consequently, with p as in (2.1),

$$m_R(x) = \int_{\mathbb{R}^n} w_R(y) p(y) e^{2\pi i y \cdot x} dy. \quad (4.9)$$

This identity follows by Fubini: w_R has compact support and p is bounded. The same overlap weight will appear in the averaged inner products in Section 5.

Define

$$\Pi_{t,R}f = \int_{\mathbb{R}^n} w_R(y)p(y)e^{2\pi it \cdot y} U_y f dy. \quad (4.10)$$

This is T_a for an integrable kernel with symbol $m_R(t + \theta)$. By (4.4) and (4.8), each $\Pi_{t,R}$ is a self-adjoint contraction.

4.4. Constructing the cutoff projections. For almost every translation parameter, the translated boundary has zero control measure. At these parameters, the approximate symbols converge in every relevant L^2 space, producing orthogonal projections and an exact formula for their differences.

Proposition 4.1. *The set*

$$\mathcal{G} = \{t \in \mathbb{R}^n : \sigma(\partial\Omega - t) = 0\} \quad (4.11)$$

is measurable and has full Lebesgue measure. For every $t \in \mathcal{G}$, $\Pi_{t,R}$ converges strongly as $R \rightarrow \infty$ to an orthogonal projection Π_t . For $s, t \in \mathcal{G}$ and $f \in \mathcal{H}$,

$$\|\Pi_t f\|^2 = \int_{\mathbb{R}^n} \mathbf{1}_\Omega(t + \theta) d\sigma_f(\theta), \quad (4.12)$$

$$\|(\Pi_t - \Pi_s)f\|^2 = \int_{\mathbb{R}^n} |\mathbf{1}_\Omega(t + \theta) - \mathbf{1}_\Omega(s + \theta)|^2 d\sigma_f(\theta). \quad (4.13)$$

Proof. The function $(t, \theta) \mapsto \mathbf{1}_{\partial\Omega}(t + \theta)$ is Borel measurable, so its integral against σ is measurable. Tonelli's theorem and the assumption $|\partial\Omega| = 0$ give

$$\int_{\mathbb{R}^n} \sigma(\partial\Omega - t) dt = \int_{\mathbb{R}^n} |\partial\Omega - \theta| d\sigma(\theta) = 0. \quad (4.14)$$

Thus $\mathcal{G}$ is conull. Fix $t \in \mathcal{G}$. By (4.6), the boundary exception is null for every σ_f . The difference norm formula for the integrable kernels gives

$$\|(\Pi_{t,R} - \Pi_{t,S})f\|^2 = \int |m_R(t + \theta) - m_S(t + \theta)|^2 d\sigma_f(\theta).$$

The bounded symbols converge in $L^2(\sigma_f)$ by dominated convergence. Hence $\Pi_{t,R}f$ is Cauchy and has a limit $\Pi_t f$. Uniform boundedness makes Π_t a bounded linear contraction. Similarly,

$$\|(\Pi_{t,R}^2 - \Pi_{t,R})f\|^2 = \int |m_R(t + \theta)^2 - m_R(t + \theta)|^2 d\sigma_f(\theta) \rightarrow 0.$$

By (3.2), $\Pi_{t,R}^2 \rightarrow \Pi_t^2$ strongly, so the preceding limit implies $\Pi_t^2 = \Pi_t$. Self-adjointness passes through inner products, so Π_t is orthogonal.

Passing to limits in the norm identities for $\Pi_{t,R}$ and $\Pi_{t,R} - \Pi_{s,R}$ proves the asserted formulas. $\square$

We call $\theta \mapsto \mathbf{1}_\Omega(t + \theta)$ the cutoff symbol of Π_t . Here t is a parameter and θ is a frequency of the stationary translation action. Formula (4.12) says that $\|\Pi_t f\|^2$ is the spectral mass of f in $\Omega - t$. Formula (4.13) will allow us to construct the two boundary limits.

5. PRESERVING THE DISTANCE BOUND

We transfer the original distance estimate to the stationary Hilbert space without losing its constant. First we construct a norm-continuous family of comparison projections M_t . We then insert stationary vectors into compactly supported functions on $\mathbb{R}^n$ and average their scalar inner products.

5.1. The covariant bump kernel. The bump projection has a continuous kernel that vanishes when its two spatial arguments are sufficiently far apart. Covariance turns this kernel into operators on $\mathcal{H}$; its projection identity and bounded support give idempotence and norm continuity, respectively.

The projection Q_Γ has the continuous kernel

$$q_\Gamma(v, w) = \sum_{\lambda \in \Gamma} b(v - \lambda) \overline{b(w - \lambda)}. \quad (5.1)$$

For fixed v at most one summand is nonzero. Thus, uniformly in Γ ,

$$|q_\Gamma(v, w)| \leq \|b\|_\infty^2, \quad q_\Gamma(v, w) = 0 \quad \text{if } |v - w| > 2r. \quad (5.2)$$

Local finiteness, weak convergence of the configurations, and continuity of b imply joint continuity in (Γ, v, w) . The kernel satisfies

$$q_{\Gamma+y}(v, w) = q_\Gamma(v - y, w - y), \quad (5.3)$$

$$q_\Gamma(w, v) = \overline{q_\Gamma(v, w)}, \quad (5.4)$$

$$\int q_\Gamma(v, u) q_\Gamma(u, w) du = q_\Gamma(v, w). \quad (5.5)$$

The last identity follows by expanding the locally finite sums and using orthonormality of the bumps. The integrals are absolutely convergent because b has compact support.

Set $k_y(\Gamma) = q_\Gamma(0, -y)$, and write B_y for multiplication by k_y on $\mathcal{H}$. The map $y \mapsto k_y$ is continuous with values in $C(X)$: joint continuity is uniform on compact subsets of $X \times \mathbb{R}^n$. In particular $y \mapsto B_y U_y f$ is continuous for every $f \in \mathcal{H}$. Moreover it vanishes for $|y| > 2r$. Define

$$M_t f = \int_{\mathbb{R}^n} e^{2\pi i t \cdot y} B_y U_y f dy. \quad (5.6)$$

This is an $\mathcal{H}$ -valued integral with norm at most $\|f\| \int \|k_y\|_\infty dy$.

Covariant operator families over point set dynamical systems are studied in [18, Sections 2–3]. The construction here uses the explicit bump kernel to obtain operators on $L^2(X, \mu)$; the next proof verifies their projection property and norm continuity directly.

Proposition 5.1. *Every M_t is an orthogonal projection. The family is Lipschitz continuous in operator norm:*

$$\|M_t - M_s\| \leq L|t - s|, \quad L = 2\pi \int_{\mathbb{R}^n} |y| \|k_y\|_\infty dy < \infty. \quad (5.7)$$

Proof. For bounded f, g , invariance of μ and the change of variables $\Gamma' = \Gamma + y$ give

$$\begin{aligned} \langle B_y U_y f, g \rangle &= \int_X q_\Gamma(0, -y) f(\Gamma + y) \overline{g(\Gamma)} d\mu(\Gamma) \\ &= \int_X q_{\Gamma'}(y, 0) f(\Gamma') \overline{g(\Gamma' - y)} d\mu(\Gamma') \\ &= \langle f, B_{-y} U_{-y} g \rangle. \end{aligned}$$

Density gives $(B_y U_y)^* = B_{-y} U_{-y}$. Taking the adjoint in (5.6) and changing y to $-y$ therefore gives $M_t^* = M_t$.

For idempotence, covariance implies $k_u(\Gamma + y) = q_\Gamma(-y, -y - u)$. First let f be bounded. Expanding M_t^2 and putting $v = y + u$ yields

$$\begin{aligned} (M_t^2 f)(\Gamma) &= \int e^{2\pi i t \cdot v} \left[\int q_\Gamma(0, -y) q_\Gamma(-y, -v) dy \right] f(\Gamma + v) dv \\ &= \int e^{2\pi i t \cdot v} q_\Gamma(0, -v) f(\Gamma + v) dv = (M_t f)(\Gamma), \end{aligned}$$

where the second equality is (5.5), with the change of variable $u = -y$. The original double integral has $|y|, |u| \leq 2r$ and uniformly bounded coefficients, so Fubini applies. The equality extends to every $f \in \mathcal{H}$ by density and boundedness. Hence $M_t^2 = M_t$, and M_t is an orthogonal projection.

Finally $|e^{2\pi i t \cdot y} - e^{2\pi i s \cdot y}| \leq 2\pi|t - s||y|$. Applying the triangle inequality to the vector integral proves (5.7). Finiteness follows from (5.2). $\square$

Norm continuity of M_t comes from the integrable, compactly supported kernel. The translation operators U_y themselves need only be strongly continuous. This distinction is essential at the boundary crossing.

5.2. Compactly supported functions from a stationary vector.

We next restrict a stationary vector to a finite box along each translation orbit and add a modulation. The normalization is chosen so that averaging the squared spatial norm recovers exactly the original Hilbert-space norm.

Fix $R > 0$, $t \in \mathbb{R}^n$, and abbreviate $C = C_R$. For $f \in \mathcal{H}$ define, for almost every Γ ,

$$a_f(\Gamma)(v) = |C|^{-1/2} \mathbf{1}_C(v) e^{-2\pi i t \cdot v} f(\Gamma - v). \quad (5.8)$$

These are compactly supported functions of $v \in \mathbb{R}^n$. Tonelli's theorem and translation invariance give

$$\int_X \|a_f(\Gamma)\|_2^2 d\mu(\Gamma) = \frac{1}{|C|} \int_C \int_X |f(\Gamma - v)|^2 d\mu(\Gamma) dv = \|f\|^2. \quad (5.9)$$

In particular $a_f(\Gamma) \in L^2(\mathbb{R}^n)$ for almost every Γ . Borel representatives of f give joint measurability, and the same identity applied to a representative that vanishes almost everywhere shows that the resulting integrals do not depend on representatives. Thus these averages are well defined despite the uncountable family of translation parameters.

5.3. Computing the two averaged inner products. The two spatial projections can now be averaged in parallel. A change to the difference variable produces the same box-overlap weight in both calculations, identifying the averages with $M_{t,R}$ and $\Pi_{t,R}$.

Let

$$M_{t,R}f = \int w_R(y) e^{2\pi i t \cdot y} B_y U_y f dy. \quad (5.10)$$

For bounded f, g , expansion of the compact spatial integrals gives

$$\begin{aligned} & \int_X \langle Q_\Gamma a_f(\Gamma), a_g(\Gamma) \rangle d\mu(\Gamma) \\ &= \frac{1}{|C|} \int_{C \times C} e^{2\pi i t \cdot (v-w)} \int_X q_\Gamma(v, w) f(\Gamma - w) \overline{g(\Gamma - v)} d\mu(\Gamma) dv dw. \end{aligned}$$

In the inner integral, make the change of variables $\Gamma' = \Gamma - v$. For $y = v - w$, covariance turns the kernel into $q_{\Gamma'}(0, -y)$, and the two functions become $f(\Gamma' + y)$ and $g(\Gamma')$. The inner integral is therefore $\langle B_y U_y f, g \rangle$. For every integrable function H of the difference variable,

$$\frac{1}{|C|} \int_{C \times C} H(v - w) dv dw = \int w_R(y) H(y) dy.$$

Consequently

$$\int_X \langle Q_\Gamma a_f, a_g \rangle d\mu = \langle M_{t,R}f, g \rangle. \quad (5.11)$$

The kernel $p(v - w)$ of P is independent of Γ . The same calculation, now with inner integral $\langle U_{v-w}f, g \rangle$, gives

$$\begin{aligned} \int_X \langle P a_f, a_g \rangle d\mu &= \int w_R(y) p(y) e^{2\pi i t \cdot y} \langle U_y f, g \rangle dy \\ &= \langle \Pi_{t,R}f, g \rangle. \end{aligned} \quad (5.12)$$

All kernels are bounded on $C \times C$. Thus the expansions for bounded f, g are justified by absolute integrability and Fubini's theorem. For general

$f, g \in \mathcal{H}$, Cauchy–Schwarz, contractivity of P, Q_Γ and (5.9) bound each averaged form by $\|f\|\|g\|$. The identities extend by density.

5.4. Applying the distance bound and passing to the limit. The averaged identities retain the original operator-norm bound because the test functions preserve the averaged squared norm. We then pass to a strong limit for the spectral cutoffs and a norm limit for the comparison operators.

Related arguments integrating frame inequalities over an invariant hull measure occur in [6, Theorem 3.12] and [7, Theorem 3.5]. Those results relate frame bounds to covolumes. Here the averaged forms (5.11)–(5.12) instead transfer the operator-norm distance bound between the two projections, with the same constant γ .

Proposition 5.2. *For every $t \in \mathcal{G}$,*

$$\|\Pi_t - M_t\| \leq \gamma < 1. \quad (5.13)$$

Proof. Subtracting (5.11) from (5.12) gives

$$\langle (\Pi_{t,R} - M_{t,R})f, g \rangle = \int_X \langle (P - Q_\Gamma)a_f(\Gamma), a_g(\Gamma) \rangle d\mu. \quad (5.14)$$

Apply (3.3) for almost every Γ , followed by Cauchy–Schwarz in X and (5.9):

$$\begin{aligned} |\langle (\Pi_{t,R} - M_{t,R})f, g \rangle| &\leq \gamma \int_X \|a_f(\Gamma)\|_2 \|a_g(\Gamma)\|_2 d\mu \\ &\leq \gamma \|f\| \|g\|. \end{aligned}$$

Thus $\|\Pi_{t,R} - M_{t,R}\| \leq \gamma$ for every t and R . By (5.2) and dominated convergence,

$$\|M_{t,R} - M_t\| \leq \int_{|y| \leq 2r} |1 - w_R(y)| \|k_y\|_\infty dy \longrightarrow 0, \quad (5.15)$$

uniformly in t . For $t \in \mathcal{G}$, Proposition 4.1 gives strong convergence $\Pi_{t,R} \rightarrow \Pi_t$. Testing the distance bound on each vector and taking the limit proves (5.13). $\square$

The finite averages $M_{t,R}$ need not be projections. Their norm limit M_t is a projection by Proposition 5.1. At a boundary parameter, the spectral cutoffs will have two different strong limits, whereas M_t has a single operator-norm limit. This gives a common comparison projection for the two sides.

6. CROSSING THE SPHERE

The analytic construction is now complete. For a ball, we choose a boundary point at which the only spectral boundary mass is at frequency zero. Approaching this point from the two sides adds that component without removing any other one, creating strictly nested limiting ranges.

We now prove Theorem 1.1. Suppose that a ball $\Omega = B(a, \rho) \subset \mathbb{R}^n$, $n \geq 2$, admits an exponential Riesz basis. The preceding sections give projections Π_t and M_t with (5.13) for $t \in \mathcal{G}$. We will construct two strong limits at a boundary point t_0 , with strictly nested ranges and both at distance at most $\gamma < 1$ from M_{t_0} .

6.1. Choosing a boundary point with no other spectral boundary mass. A sphere has surface-measure-zero intersection with each of its nontrivial translates. Tonelli's theorem reverses the roles of boundary points and frequencies, yielding a point whose translated boundary has no control-measure mass away from zero.

Write $S = \partial\Omega$ and let $\nu = \mathcal{H}^{n-1}|_S$ be surface measure on S . The measure ν is finite and nonzero.

Lemma 6.1. *For every nonzero $\theta \in \mathbb{R}^n$,*

$$\nu(S \cap (S - \theta)) = 0. \quad (6.1)$$

Moreover, S has Lebesgue measure zero.

Proof. If $x \in S \cap (S - \theta)$, then $|x - a|^2 = \rho^2$ and $|x + \theta - a|^2 = \rho^2$. Subtraction gives

$$2(x - a) \cdot \theta = -|\theta|^2.$$

For $\theta \neq 0$, this is a hyperplane. Its intersection with the sphere is empty, a point, or an $(n - 2)$ -dimensional sphere, hence has zero $\mathcal{H}^{n-1}$ -measure. Finally a sphere in $\mathbb{R}^n$ has zero n -dimensional Lebesgue measure. $\square$

We need the reverse viewpoint: fix a boundary point t_0 and control the frequencies θ for which $t_0 + \theta \in S$. Tonelli's theorem interchanges these roles. Using the control measure σ from (4.5), we obtain

$$\begin{aligned} \int_S \sigma((S - t) \setminus \{0\}) d\nu(t) &= \int_{\mathbb{R}^n \setminus \{0\}} \nu(S \cap (S - \theta)) d\sigma(\theta) \\ &= 0. \end{aligned} \quad (6.2)$$

The integrand on the left is nonnegative and $\nu(S) > 0$. Thus we can choose $t_0 \in S$ such that

$$\sigma((S - t_0) \setminus \{0\}) = 0. \quad (6.3)$$

By (4.6), the same set is null for every σ_f . This is why we introduced one control measure: the chosen point works for every vector in $\mathcal{H}$ simultaneously.

Zero must be excluded from this null-set statement. Indeed, $\sigma_{\mathbf{1}} = \delta_0$, so the constant vector has spectral mass there. This is precisely the mass that will enter the cutoff.

6.2. Approaching the point from outside and inside. The cutoff projection need not be defined at the chosen boundary point. We therefore approach it through good parameters in ambient open sets on the two sides, rather than along a prescribed line. The limiting symbols agree everywhere except at the zero-frequency atom, up to a common spectral null set.

The good set $\mathcal{G}$ has full Lebesgue measure in $\mathbb{R}^n$, but this does not imply that it contains t_0 . In fact $t_0 \notin \mathcal{G}$: since $0 \in S - t_0$ and $\sigma_1 = \delta_0$, condition (4.11) cannot hold there. We therefore approach t_0 through good parameters and take limits of their projections.

Every neighborhood of a point on the sphere meets both Ω and its strict exterior in nonempty open sets. Each has positive Lebesgue measure, so each meets $\mathcal{G}$. We can consequently choose

$$t_j^- \in \mathcal{G} \cap (\mathbb{R}^n \setminus \overline{\Omega}), \quad t_j^+ \in \mathcal{G} \cap \Omega, \quad t_j^\pm \longrightarrow t_0.$$

These choices use open neighborhoods in the ambient space. There is no need to restrict the good parameters to a prescribed radial line.

Write the corresponding cutoff symbols as

$$a_j^\pm(\theta) = \mathbf{1}_\Omega(t_j^\pm + \theta).$$

If $t_0 + \theta$ is away from S , membership in Ω is locally constant, so both symbols eventually equal $\mathbf{1}_\Omega(t_0 + \theta)$. At $\theta = 0$, the exterior symbol is always zero and the interior symbol is always one. All remaining cases lie in $(S - t_0) \setminus \{0\}$, which is null for every σ_f by (6.3) and (4.6). Thus, for every $f \in \mathcal{H}$, we have $a_j^\pm \rightarrow m_\pm$ σ_f -almost everywhere, where

$$m_-(\theta) = \mathbf{1}_{\Omega - t_0}(\theta), \tag{6.4}$$

$$m_+(\theta) = \mathbf{1}_{\Omega - t_0}(\theta) + \mathbf{1}_{\{0\}}(\theta). \tag{6.5}$$

Since Ω is open and $t_0 \in S$, zero does not belong to $\Omega - t_0$. Both limits are therefore indicator functions, and $m_- \leq m_+$. The only added spectral component is at zero.

6.3. Turning the limiting symbols into projections. The exact difference formula converts convergence of the indicator functions into strong convergence of the projections. Their limiting norm identities will then allow us to read inclusion of the ranges directly from inclusion of the spectral sets.

For either choice of sign, dominated convergence gives $a_j^\pm \rightarrow m_\pm$ in $L^2(\sigma_f)$ for every $f \in \mathcal{H}$. Formula (4.13) therefore yields

$$\|(\Pi_{t_j^\pm} - \Pi_{t_k^\pm})f\| = \|a_j^\pm - a_k^\pm\|_{L^2(\sigma_f)} \longrightarrow 0.$$

Thus $(\Pi_{t_j^\pm} f)_j$ is Cauchy for every f . Since the projections are contractions, their limits define bounded linear operators R_-, R_+ with

$$\Pi_{t_j^-} \longrightarrow R_-, \quad \Pi_{t_j^+} \longrightarrow R_+ \quad \text{strongly.} \tag{6.6}$$

By the strong-convergence facts recorded in Section 3, R_- and R_+ are orthogonal projections. Passing to the limit in (4.12) gives

$$\|R_{\pm}f\|^2 = \int_{\mathbb{R}^n} m_{\pm}(\theta) d\sigma_f(\theta). \quad (6.7)$$

This formula lets us read the order of the limiting projections from the order of their symbols.

6.4. Strict inclusion of subspaces and the contradiction. Order of the limiting symbols gives inclusion of the ranges, and the constant vector makes this inclusion strict. Norm continuity of the comparison family then places both limiting projections within distance less than one of the same projection, contradicting the elementary projection lemma.

Since $m_- \leq m_+$, (6.7) implies $\|R_-f\| \leq \|R_+f\|$ for every f . In particular, $R_+f = 0$ implies $R_-f = 0$. Thus

$$\ker R_+ \subset \ker R_-, \quad \text{Ran } R_- \subset \text{Ran } R_+,$$

where the second inclusion follows by taking orthogonal complements.

To see that this inclusion is strict, use the constant vector. Its spectral measure is δ_0 , so (6.7) gives

$$R_- \mathbf{1} = 0, \quad \|R_+ \mathbf{1}\| = \|\mathbf{1}\| = 1.$$

By Pythagoras, an orthogonal projection preserves a vector's norm only when it fixes that vector. Hence $R_+ \mathbf{1} = \mathbf{1}$: the constant function belongs to the larger range and is orthogonal to the smaller one. We have proved

$$\text{Ran } R_- \subsetneq \text{Ran } R_+. \quad (6.8)$$

More precisely, the difference is an orthogonal projection, and

$$\|(R_+ - R_-)f\|^2 = \sigma_f(\{0\}), \quad (R_+ - R_-)\mathbf{1} = \mathbf{1}. \quad (6.9)$$

The jump need not have rank one; the constant vector only ensures that it is nonzero.

Both limits also inherit a common distance bound. By Proposition 5.1, $M_{t_j^{\pm}} \rightarrow M_{t_0}$ in operator norm. For every $f \in \mathcal{H}$,

$$\begin{aligned} \|(R_{\pm} - M_{t_0})f\| &= \lim_j \|(\Pi_{t_j^{\pm}} - M_{t_0})f\| \\ &\leq \limsup_j \left(\gamma + \|M_{t_j^{\pm}} - M_{t_0}\| \right) \|f\| = \gamma \|f\|. \end{aligned}$$

Consequently

$$\|R_- - M_{t_0}\| \leq \gamma < 1, \quad \|R_+ - M_{t_0}\| \leq \gamma < 1. \quad (6.10)$$

This comparison is valid even though $t_0 \notin \mathcal{G}$: M_t is defined and is a projection for every $t \in \mathbb{R}^n$. Lemma 2.3 contradicts (6.8) and (6.10). This proves Theorem 1.1.

Remark 6.2 (The role of the stationary space). The invariant probability measure supplies a nonzero constant vector with spectral measure δ_0 . The ordinary translation representation $(V_y a)(v) = a(v - y)$ on $L^2(\mathbb{R}^n)$ has no nonzero invariant vector. With our Fourier convention, its scalar spectral measures are $|\mathcal{F}a(\theta)|^2 d\theta$, so changing a cutoff at the single point zero has no effect on that space. On $\mathcal{H}$, the same change gives $R_- \mathbf{1} = 0$ and $R_+ \mathbf{1} = \mathbf{1}$. The averaging argument brings both this invariant vector and the original distance bound into the same stationary space.

Remark 6.3. A sphere has no overlap of positive surface measure with a nontrivial translate. This fails for the one-dimensional interval: a translation can send one endpoint to the other. It also fails for opposite sides of a rectangle. A boundary crossing can then exchange spectral components, and the contradiction from strict inclusion does not follow.

Remark 6.4 (A more general boundary condition). Let $\Omega \subset \mathbb{R}^n$ be a nonempty bounded open set with $|\partial\Omega| = 0$. Suppose its boundary carries a finite nonzero positive Borel measure ν such that

$$\nu(\partial\Omega \cap (\partial\Omega - \theta)) = 0 \quad (\theta \neq 0).$$

Assume also that, for ν -almost every $t \in \partial\Omega$, every neighborhood of t meets both Ω and $\mathbb{R}^n \setminus \overline{\Omega}$ in sets of positive Lebesgue measure. Then $L^2(\Omega)$ admits no Riesz basis of exponentials. Indeed, Sections 2–5 apply under these hypotheses. Tonelli’s theorem, applied with ν , supplies a point satisfying both the spectral boundary-null condition and the stated two-sided neighborhood condition. The latter provides the two sequences of good parameters. Their symbols have exactly the limits (6.4)–(6.5), and the same arguments using strong limits and norm continuity give a contradiction. In particular one may use surface measure whenever it is finite and nonzero and satisfies the displayed overlap condition and the two-sided neighborhood condition.

7. TRIANGLES

For a triangle, tangential translations can produce overlap of positive length along an edge, so the sphere argument does not apply verbatim. We instead isolate one edge from the other two. Crossing it then brings all the surviving spectral boundary mass into the cutoff at once, again giving strictly nested limiting ranges.

Theorem 7.1. *No nondegenerate triangle in $\mathbb{R}^2$ admits a Riesz basis of exponentials.*

Proof. An invertible affine change of variables preserves the Riesz basis property. Indeed, the pullback under $x = a + Ay$ sends e_λ to $e^{2\pi i \lambda \cdot a} e_{A^T \lambda}$

and multiplies the L^2 norm by $|\det A|^{-1/2}$. The phase factors have modulus one, and the frequency map is invertible. Thus we may assume

$$\Omega = \{x_1 > 0, x_2 > 0, x_1 + x_2 < 1\}.$$

Suppose it has an exponential Riesz basis. Since Ω is bounded and its boundary has area zero, Sections 2–5 give the control measure σ , the conull set $\mathcal{G}$, and the projections with (5.13).

We first choose a point in the interior of the horizontal edge such that the other two edges carry no translated spectral mass. Their nonparallel supporting lines make the required Tonelli argument possible.

Let $L = \{(a, 0) : 0 < a < 1\}$ and $F = \partial\Omega \setminus L$. The set F consists of the two other closed edges, including all vertices. For $\nu = \mathcal{H}^1|_L$ and every $\theta \in \mathbb{R}^2$,

$$\nu(L \cap (F - \theta)) = 0,$$

because a horizontal segment meets a translated vertical or diagonal supporting line in at most one point. Tonelli's theorem therefore gives

$$\int_L \sigma(F - t) d\nu(t) = \int_{\mathbb{R}^2} \nu(L \cap (F - \theta)) d\sigma(\theta) = 0.$$

Choose $t_0 \in L$ with $\sigma(F - t_0) = 0$. This set is null for every σ_f by (4.6). Once the other edges have been removed as spectral null sets, the sign of one normal coordinate controls every surviving boundary point. Good parameters approaching from opposite sides therefore have ordered limiting symbols.

Near t_0 , the triangle coincides with the half-plane $x_2 > 0$. Choose good parameters $t_j^+ \in \mathcal{G} \cap \Omega$ and $t_j^- \in \mathcal{G} \cap (\mathbb{R}^2 \setminus \overline{\Omega})$ tending to t_0 inside such a neighborhood. Their second coordinates are positive and negative, respectively. For each point $t_0 + \theta \in L$, membership in the triangle is likewise determined locally by positivity of the second coordinate. Therefore $\mathbf{1}_\Omega(t_j^+ + \theta) \rightarrow 1$ and $\mathbf{1}_\Omega(t_j^- + \theta) \rightarrow 0$ on $L - t_0$. Away from the boundary both symbols converge to $\mathbf{1}_\Omega(t_0 + \theta)$. All other boundary points belong to the common null set $F - t_0$.

The limiting symbols are consequently

$$m_- = \mathbf{1}_{\Omega - t_0}, \quad m_+ = \mathbf{1}_{\Omega - t_0} + \mathbf{1}_{L - t_0}.$$

They are indicators of nested sets.

The remaining steps are the same as for the ball: the difference formula gives strong limits, the constant vector makes their ranges strictly nested, and norm continuity supplies a common nearby projection.

Formula (4.13) and dominated convergence make $(\Pi_{t_j^\pm} f)_j$ Cauchy for every f . As in Section 6, the resulting strong limits R_-, R_+ are orthogonal projections with

$$\|R_\pm f\|^2 = \int m_\pm d\sigma_f.$$

Thus $\|R_-f\| \leq \|R_+f\|$ for every f , which gives $\text{Ran } R_- \subset \text{Ran } R_+$ by taking kernels and orthogonal complements. Since $0 \in L - t_0$ and $\sigma_1 = \delta_0$, we have $R_-1 = 0$ and $R_+1 = 1$; the inclusion is strict. Finally $M_{t_j^\pm} \rightarrow M_{t_0}$ in norm, so (5.13) gives $\|R_\pm - M_{t_0}\| \leq \gamma < 1$, contradicting Lemma 2.3.

In this case, $\|(R_+ - R_-)f\|^2 = \sigma_f(L - t_0)$. The jump may contain nonzero tangential frequencies, but the constant vector still guarantees that the jump is nonzero. $\square$

Remark 7.2 (Convex polygons with an unpaired edge). The same proof excludes an exponential Riesz basis on the interior of a convex polygon having an edge with no parallel partner. Here edges are maximal boundary segments. In particular the conclusion holds for every convex polygon with an odd number of sides: a bounded convex polygon has at most two maximal edges in each unoriented direction, so an odd number of sides cannot be partitioned into parallel pairs.

To verify the extension, let L be the relative interior of an unpaired edge and put $F = \partial\Omega \setminus L$. Every other edge is nonparallel to L , so for $\nu = \mathcal{H}^1|_L$,

$$\nu(L \cap (F - \theta)) = 0 \quad (\theta \in \mathbb{R}^2).$$

The intersections with the translated supporting lines are finite; the two endpoints contribute only finitely many further points. Tonelli's theorem gives $t_0 \in L$ with $\sigma(F - t_0) = 0$. Fix the inward unit normal to L . At each point of L , membership in Ω is locally determined by positive displacement in this normal direction. Choose good parameters approaching t_0 from the two local half-planes. For every $t_0 + \theta \in L$, the same normal sign determines their eventual membership. The symbols are therefore the same m_-, m_+ as in the triangle proof, with this edge L . Their limiting projections satisfy $\text{Ran } R_- \subsetneq \text{Ran } R_+$ because $0 \in L - t_0$. Norm continuity of M_t again supplies the common comparison projection M_{t_0} and Lemma 2.3 gives the contradiction.

8. BOUNDED CONVEX DOMAINS WITH C^2 BOUNDARY

The boundary criterion extends the nonexistence result from balls to all nonempty bounded convex open sets with C^2 boundary. The key is to use surface measure on a small positively curved patch, rather than on the whole boundary, which may contain flat pieces. We first find such a patch and then show that its intersection with every nontrivial translate of the boundary has surface measure zero.

Corollary 8.1. *Let $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be a nonempty bounded convex open set with C^2 boundary. Then $L^2(\Omega)$ admits no Riesz basis of exponentials.*

Neither strict convexity of the whole domain nor positive curvature at every boundary point is required.

A ball containing the closure of the domain touches its boundary at a point where all principal curvatures are positive. Continuity gives a patch with the same property. Convexity then makes each supporting hyperplane along this patch meet the closed domain at only one point.

Lemma 8.2. *Under the hypotheses of Corollary 8.1, put $K = \overline{\Omega}$ and $S = \partial\Omega$. There is a nonempty relatively open patch $U \subset S$ on which the second fundamental form is positive definite, with the convention that convex principal curvatures are nonnegative. Moreover, if $N(x)$ denotes the outward unit normal, then*

$$K \cap \{z : N(x) \cdot (z - x) = 0\} = \{x\} \quad (x \in U). \quad (8.1)$$

Proof. Choose $c \in \Omega$ and let

$$R = \max_{z \in K} |z - c| > 0.$$

Then $K \subset \overline{B}(c, R)$, with contact at some $p \in S$. The tangent hyperplane to the ball at p supports K and, since S is C^2 , is also the tangent hyperplane to S . After a translation and rotation, we may assume that

$$p = 0, \quad c = -Re_n, \quad N(p) = e_n.$$

Locally write S as $x_n = f(x')$, with $f \in C^2$, $f(0) = 0$ and $Df(0) = 0$. Containment in the ball gives

$$f(x') \leq -R + \sqrt{R^2 - |x'|^2},$$

with equality at $x' = 0$. Taking second derivatives at this contact point in the sense of quadratic forms yields

$$D^2f(0) \leq -\frac{1}{R}I.$$

Thus the second fundamental form at p , represented in these coordinates by $-D^2f(0)$, is positive definite. By continuity it remains positive definite on a sufficiently small relatively open patch U .

To prove (8.1), fix $x \in U$ and suppose that another point $z \in K$ lies in the supporting hyperplane at x . Convexity places the segment $[x, z]$ in K and in that hyperplane, hence in S . In coordinates centered at x with horizontal tangent hyperplane, the boundary is the graph of a function g with $g(0) = 0$, $Dg(0) = 0$ and $D^2g(0)$ negative definite. The segment would give $g(tv) = 0$ for a nonzero tangent vector v and all sufficiently small $t > 0$, contradicting

$$g(tv) = \frac{t^2}{2} D^2g(0)[v, v] + o(t^2) < 0.$$

This proves the singleton supporting-face property. $\square$

We now compare the outward normals at two boundary points separated by a fixed translation. Nonparallel normals give a transverse intersection of codimension two. Equal normals are ruled out by the

singleton supporting-face property, while opposite normals force the entire intersection to consist of one point.

Lemma 8.3. *Let U be the patch in Lemma 8.2. Then*

$$\mathcal{H}^{n-1}(U \cap (S - \theta)) = 0 \quad (\theta \in \mathbb{R}^n \setminus \{0\}). \quad (8.2)$$

Proof. Fix $\theta \neq 0$. At any $x \in U \cap (S - \theta)$, put $y = x + \theta \in S$. The normals to S and $S - \theta$ at x are $N(x)$ and $N(y)$, respectively.

If these normals are not parallel, the implicit function theorem makes the intersection locally a C^1 submanifold of dimension $n - 2$. The set of all such transverse intersection points is covered by countably many of these neighborhoods, since $\mathbb{R}^n$ is second countable. It therefore has zero $\mathcal{H}^{n-1}$ -measure.

The equality $N(y) = N(x)$ is impossible. Indeed, x and y would maximize the same linear functional on K and so lie in the same supporting hyperplane, contrary to (8.1) and $x \neq y$.

Finally, suppose $N(y) = -N(x)$ and write $v = N(x)$. The supporting inequalities at x and y give

$$K \subset \{z : v \cdot (z - x) \leq 0\}, \quad K - \theta \subset \{z : v \cdot (z - x) \geq 0\}.$$

Consequently, (8.1) implies

$$K \cap (K - \theta) \subset K \cap \{z : v \cdot (z - x) = 0\} = \{x\}.$$

Since x belongs to both boundaries, $S \cap (S - \theta) = \{x\}$ in this case. A singleton has zero $\mathcal{H}^{n-1}$ -measure because $n \geq 2$. If this case does not occur, every intersection point in U is transverse. This proves (8.2). $\square$

Surface measure restricted to the patch now satisfies the non-overlap hypothesis of Remark 6.4. The remaining hypotheses follow from boundary regularity and convexity, so the analytic construction and the final projection contradiction need no changes.

Proof of Corollary 8.1. The compact C^2 hypersurface S has zero n -dimensional Lebesgue measure and finite $(n - 1)$ -dimensional surface measure. Since U is a nonempty relatively open patch, the positive Borel measure

$$\nu = \mathcal{H}^{n-1}|_U$$

is finite and nonzero. Lemma 8.3 gives

$$\nu(S \cap (S - \theta)) = \mathcal{H}^{n-1}(U \cap (S - \theta)) = 0 \quad (\theta \neq 0).$$

Every neighborhood of a point $t \in S$ meets Ω in a nonempty open set, because t is a boundary point. It also contains a nonempty open subset of the strict exterior: the supporting inequality $N(t) \cdot (z - t) \leq 0$ holds throughout K , so points sufficiently close to t on the other side of this hyperplane lie outside K . Both intersections therefore have positive Lebesgue measure. All the hypotheses of Remark 6.4 hold, and its conclusion proves the corollary. $\square$

9. COMPLETE AND MINIMAL EXPONENTIALS ON BOUNDED CONVEX DOMAINS

For domains already known to admit exponential Riesz bases, existence of a complete minimal system follows at once; this includes the convex polytopes with centrally symmetric faces treated in [5]. The theorem below imposes no such geometric restriction.

9.1. Statement and Fourier conventions. A family of function in $L^2(\Omega)$ is *minimal* when no member belongs to the closed span of the others. Equivalently, it admits individual biorthogonal functions; no uniform bound on their norms is required.

Theorem 9.1. *Let $n \geq 1$ and let $\Omega \subset \mathbb{R}^n$ be nonempty, bounded, open, and convex. There is a locally finite sequence $\Lambda \subset \mathbb{R}^n$ such that $\{e_\lambda : \lambda \in \Lambda\}$ is complete and minimal in $L^2(\Omega)$. Its biorthogonal functions are supported in the John ellipsoid of $\overline{\Omega}$.*

The statement concerns ordinary minimality. It does not assert a Riesz basis, a Schauder basis, or completeness of the biorthogonal family in $L^2(\Omega)$.

For $n = 1$, the domain is an interval (a, b) and the frequencies $(b - a)^{-1}\mathbb{Z}$ give an orthogonal basis. We therefore assume $n \geq 2$ throughout the construction below. No smoothness or strict convexity of the boundary is assumed.

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot z} dx, \quad z \in \mathbb{C}^n,$$

for compactly supported integrable functions, and its distributional extension. Two classical Fourier facts will be used explicitly. First, an entire function on $\mathbb{C}^n$ of finite exponential type which is polynomially bounded on $\mathbb{R}^n$ is the Fourier transform of a compactly supported distribution. This is the Paley–Wiener–Schwartz theorem; this formulation is recalled in [27, Introduction, pp. 1–2]. Second, for nonzero compactly supported distributions u, v ,

$$\text{conv supp}(u * v) = \text{conv supp } u + \text{conv supp } v. \quad (9.1)$$

This is the Titchmarsh–Lions convolution support theorem; see [11, Theorem 4.2]. Standard Bessel formulas are taken from [24].

9.2. Normalization by the John ellipsoid. Write $B = \{x \in \mathbb{R}^n : |x| < 1\}$ and $v_n = |B|$. By John's theorem, the convex body $\bar{\Omega}$ has a unique inscribed ellipsoid of maximal volume, its *John ellipsoid*; see [1, Theorem 3.1, p. 13]. Normalizing this ellipsoid to $\bar{B}$, uniqueness gives the following statement.

Lemma 9.2. *After an invertible affine change of variables, $\bar{B}$ is the John ellipsoid of $\bar{\Omega}$ and*

$$B \subset \Omega, \quad \{t \in \mathbb{R}^n : \bar{B} + t \subset \bar{\Omega}\} = \{0\}. \quad (9.2)$$

We henceforth work in this normalization. Denote by PW_Ω the Fourier image of $L^2(\Omega)$, with physical-space functions extended by zero. Every member is entire of finite exponential type. It and its first derivatives are bounded on $\mathbb{R}^n$, since Ω is bounded and $L^2(\Omega) \subset L^1(\Omega)$.

9.3. Spheres and finite-dimensional interpolation spaces. Set $\nu = n/2 - 1$ and $\beta = (n+1)/2$. Let a_j , $j \geq 1$, be the increasing positive zeros of $J_{\nu+1} = J_{n/2}$. They are simple, tend to infinity, and $J_\nu(a_j) \neq 0$. The Fourier transform of the ball gives

$$\begin{aligned} \kappa(z) &= \frac{\widehat{\mathbf{1}_B}(z)}{v_n}, & \kappa(\xi) &= k(2\pi|\xi|), \\ k(s) &= 2^{\nu+1}\Gamma(\nu+2)\frac{J_{\nu+1}(s)}{s^{\nu+1}}, & q(z) &= 4\pi^2 \sum_{k=1}^n z_k^2. \end{aligned} \quad (9.3)$$

Here $k(0) = 1$, and $\mathcal{P}_m$ denotes complex polynomials of total degree at most m in n variables. The Bessel quotient defining k is understood by its even power series; thus k is entire, also when n is odd. The same series defines κ on $\mathbb{C}^n$ without choosing a square root. In q the squares are complex squares, with no conjugation. Define

$$Q_N(z) = \prod_{j=1}^N (q(z) - a_j^2), \quad Q_0 = 1, \quad S_j = \{\xi \in \mathbb{R}^n : 2\pi|\xi| = a_j\},$$

and $V_N = \{P\kappa/Q_N : P \in \mathcal{P}_{2N}\}$. The Bessel product gives

$$\kappa(z) = \prod_{j=1}^{\infty} \left(1 - \frac{q(z)}{a_j^2}\right), \quad (9.4)$$

locally uniformly on $\mathbb{C}^n$.

Lemma 9.3. *The spaces V_N are contained in $PW_B \subset PW_\Omega$. They are nested, and*

$$\dim V_N = \dim \mathcal{P}_{2N} = \binom{2N+n}{n}.$$

The function κ/Q_N is nonzero at every point of $\{0\} \cup \bigcup_{j \leq N} S_j$, and every member of V_N vanishes on S_j for $j > N$.

Proof. For a positive zero a of $J_{\nu+1}$ define

$$\phi_a(x) = \frac{1}{v_n a^2} \left(\frac{|x|^{-\nu} J_\nu(a|x|)}{J_\nu(a)} - 1 \right) \mathbf{1}_B(x).$$

The value at the origin is defined by continuity; the interior expression is smooth there by its power series in $|x|^2$. Both its value and its normal derivative vanish on ∂B , since

$$\frac{d}{dr} (r^{-\nu} J_\nu(ar)) = -ar^{-\nu} J_{\nu+1}(ar).$$

The Bessel equation therefore gives, including the zero extension,

$$(-\Delta - a^2)\phi_a = \mathbf{1}_B/v_n.$$

There are no boundary distributions in this equation. Fourier transformation shows that $\widehat{\phi_a} = \kappa/(q - a^2)$, with removable singularities. Partial fractions in q now show that κ/Q_N has a compactly supported inverse transform in $\overline{B}$.

The real-axis Bessel asymptotic gives

$$\kappa(\xi)/Q_N(\xi) = O(|\xi|^{-2N-\beta}) \quad (|\xi| \rightarrow \infty). \quad (9.5)$$

Thus $P\kappa/Q_N \in L^2(\mathbb{R}^n)$ for $\deg P \leq 2N$: the radial majorant for its squared norm at infinity is $r^{n-1-2\beta} = r^{-2}$. Its inverse distribution is obtained by differentiating a distribution supported in $\overline{B}$, hence has the same support inclusion. Plancherel makes it an L^2 function supported there. Since ∂B has Lebesgue measure zero, this proves membership in PW_B , including $N = 0$ directly.

Nesting follows by multiplying the numerator by $q - a_{N+1}^2$. Multiplication by κ/Q_N , nonzero on an open set, is injective on polynomials. The dimension and the remaining assertions follow from the simple zeros and (9.4). $\square$

The spaces V_N form a nested finite-dimensional interpolation hierarchy, analogous to a multiresolution decomposition. Here each level adds polynomial data on one further Bessel-zero sphere.

9.4. Division and the support of the quotient. We first pass from real spheres to their complex zero hypersurfaces. This step is valid in every dimension $n \geq 2$.

Lemma 9.4. *Let $r > 0$. If an entire function F on $\mathbb{C}^n$ vanishes on $\{\xi \in \mathbb{R}^n : |\xi| = r\}$, then it vanishes on the complex quadric*

$$\mathcal{Q}_r = \left\{ z \in \mathbb{C}^n : \sum_{k=1}^n z_k^2 = r^2 \right\}.$$

It is locally divisible by $\sum_{k=1}^n z_k^2 - r^2$ at every point of this quadric.

Proof. For $z = x + iy \in \mathcal{Q}_r$, with $x, y \in \mathbb{R}^n$, one has $x \cdot y = 0$ and $|x|^2 - |y|^2 = r^2$. In particular $x \neq 0$. There is a real two-dimensional linear subspace E containing x and y ; if $y = 0$, choose any such subspace containing x . In orthonormal coordinates on its complexification, the quadric section is parametrized by

$$z_1 = \frac{r}{2}(w + w^{-1}), \quad z_2 = \frac{r}{2i}(w - w^{-1}), \quad w \in \mathbb{C} \setminus \{0\}.$$

This parametrization is onto: its inverse is $w = (z_1 + iz_2)/r$, which is nonzero because $(z_1 + iz_2)(z_1 - iz_2) = r^2$. The unit circle in the w plane maps onto the real circle $E \cap rS^{n-1}$. The one-variable identity theorem therefore shows that F vanishes on the whole section, in particular at z . Finally, the gradient of the defining polynomial is $2z \neq 0$ on $\mathcal{Q}_r$. Local holomorphic coordinates make that polynomial one coordinate; vanishing when this coordinate is zero gives the asserted local divisibility. $\square$

We next prove the growth estimates needed for division. Entireness of a quotient alone does not give compact support of its inverse transform.

Lemma 9.5. *Suppose A, D are entire on $\mathbb{C}^n$, $|A(z)| + |D(z)| \leq Ce^{R|z|}$, and $D(0) \neq 0$. If A/D extends to an entire function, that quotient has finite exponential type.*

Proof. Divide both A and D by $D(0)$, adjusting constants, so $D(0) = 1$. For a complex unit vector u , restrict the functions to the line $w \mapsto wu$. Jensen's formula gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log |D(re^{it}u)| dt \geq 0.$$

Hence the mean of $\log^- |D|$ on that circle is at most the mean of $\log^+ |D|$. Writing $G = A/D$, we obtain, uniformly in u ,

$$\frac{1}{2\pi} \int_0^{2\pi} \log^+ |G(re^{it}u)| dt \leq 2 \log^+ C + 2Rr.$$

The Poisson inequality for the subharmonic function $\log |G(wu)|$, using a circle of radius $2|w|$ for $|w| \geq 1$, bounds its value by three times the last mean. This gives $|G(z)| \leq C'e^{R'|z|}$ for $|z| \geq 1$, uniformly in the direction u ; the compact remainder is harmless. Radii passing through zeros are handled by limits. $\square$

Lemma 9.6. *If $F \in PW_\Omega$ vanishes on every S_j , $j > N$, then*

$$G = Q_N F / \kappa \tag{9.6}$$

extends to an entire function which is the Fourier transform of a compactly supported distribution.

Proof. By Lemma 9.4, the numerator $Q_N F$ vanishes on all the complex quadrics $\mathcal{Q}_r$ for $r = a_j/(2\pi)$: for $j \leq N$ the corresponding factor is in Q_N , and for $j > N$ use the hypothesis. They are exactly the zero hypersurfaces of κ in (9.4), each of multiplicity one, and they do not intersect each other. They form a locally finite collection because $a_j \rightarrow \infty$. Local divisibility therefore removes all singularities of (9.6).

The numerator $Q_N F$ and the denominator κ have finite exponential type. Since $\kappa(0) = 1$, Lemma 9.5 gives the same property for G . We next prove that G is polynomially bounded on $\mathbb{R}^n$.

The standard Bessel asymptotics and their differentiated versions give, for sufficiently large s ,

$$|k(s)| \geq c(1+s)^{-\beta} \min\{1, \text{dist}(s, \{a_j : j \geq 1\})\}. \quad (9.7)$$

For completeness, with $c_n > 0$ and $\gamma_n = \pi(n+1)/4$,

$$\begin{aligned} k(s) &= c_n s^{-\beta} (\cos(s - \gamma_n) + O(s^{-1})), \\ k'(s) &= c_n s^{-\beta} (-\sin(s - \gamma_n) + O(s^{-1})). \end{aligned}$$

The zeros satisfy $a_j = (j + (n-1)/4)\pi + O(j^{-1})$. Off fixed small neighborhoods of these zeros the cosine is bounded away from zero. Inside each neighborhood the derivative has constant sign and magnitude at least $cs^{-\beta}$; integration from the zero proves (9.7).

For $\theta \in S^{n-1}$, set

$$H_\theta(s) = \prod_{j \leq N} (s^2 - a_j^2) F(s\theta/(2\pi)).$$

Uniform boundedness of F and its first real derivatives gives

$$|H_\theta(s)| + |H'_\theta(s)| \leq C(1+s)^{2N} \quad (s \geq 0),$$

uniformly in θ . Also $H_\theta(a_j) = 0$ for every j . Near a zero, integrate this derivative estimate from that zero; away from the zeros use the bound on H_θ itself. Combining with (9.7) proves

$$|G(\xi)| \leq C(1+|\xi|)^{2N+\beta} \quad (\xi \in \mathbb{R}^n).$$

The bounded real-frequency region is covered by continuity. The stated Paley–Wiener–Schwartz theorem now gives a compactly supported distribution with Fourier transform G . $\square$

Proposition 9.7. *For a domain satisfying (9.2),*

$$\{F \in PW_\Omega : F|_{S_j} = 0 \text{ for all } j > N\} = V_N. \quad (9.8)$$

Proof. Let $F = \widehat{f}$ belong to the left side, and suppose $F \neq 0$. By Lemma 9.6, $G = Q_N F / \kappa = \widehat{g}$ for a nonzero compactly supported distribution g . Put $u = \mathbf{1}_B / v_n$. The product identity $\kappa G = Q_N F$ gives

$$u * g = \prod_{j \leq N} (-\Delta - a_j^2) f. \quad (9.9)$$

The right side is supported in $\overline{\Omega}$. The convolution support theorem (9.1) implies

$$\overline{B} + \text{conv supp } g = \text{conv supp}(u * g) \subset \overline{\Omega}.$$

Condition (9.2) forces $\text{supp } g \subset \{0\}$. A distribution supported at one point is a finite linear combination of derivatives of its Dirac measure. Hence $G = P$ is a polynomial, and

$$F = P\kappa/Q_N.$$

We check the degree rather than assuming it from support. If $m = \deg P$, its leading homogeneous part has absolute value bounded below on some open patch of S^{n-1} . Uniformly on a smaller patch, $|P(r\theta)| \geq cr^m$ for large r . On real radial intervals of fixed positive length repeated at bounded spacing, the Bessel asymptotic gives $|\kappa(r\theta)| \geq cr^{-\beta}$. Also $|Q_N(r\theta)| \leq Cr^{2N}$ for large r . Thus the polar integral of $|F|^2$ on this patch and these radial intervals is bounded below by a constant times the integral of

$$r^{2m-4N-2\beta+n-1} = r^{2m-4N-2}.$$

If $m \geq 2N + 1$, this diverges. Since $F \in L^2(\mathbb{R}^n)$, it follows that $m \leq 2N$. In particular, the degree bound is the same in every dimension. Therefore $F \in V_N$. The zero function is immediate, and the reverse inclusion is Lemma 9.3. $\square$

9.5. Selection of the frequencies. Call a finite set

$$A \subset \{0\} \cup \bigcup_{j \leq m} S_j$$

an interpolation set at level m if evaluation on A maps $\mathcal{P}_{2m}$ isomorphically onto $\mathbb{C}^A$. By Lemma 9.3, this is equivalent to interpolation by V_m . At level zero take $A = \{0\}$.

Lemma 9.8. *For such a set A , the exponentials with frequencies in $A \cup \bigcup_{j > m} S_j$ have dense span in $L^2(\Omega)$.*

Proof. The Fourier transform of a function orthogonal to all those exponentials belongs to V_m by Proposition 9.7. Its vanishing on A and interpolation then force it to be zero. $\square$

Lemma 9.9. *Let A be an interpolation set at level m , and let C be a finite set of distinct points on spheres with index greater than m . There are $N > m$ and an interpolation set A' at level N such that*

$$A \cup C \subset A', \quad A' \setminus A \subset \bigcup_{m < j \leq N} S_j.$$

Proof. Choose $N > m$ containing all sphere indices occurring in C and satisfying $2N \geq |A \cup C| - 1$. Evaluations at these distinct points are

independent on $\mathcal{P}_{2N}$: for each $\lambda \in A \cup C$, the polynomial

$$\prod_{\substack{\mu \in A \cup C \\ \mu \neq \lambda}} \frac{(\xi - \mu) \cdot (\lambda - \mu)}{|\lambda - \mu|^2}$$

has the required Kronecker values and degree $|A \cup C| - 1$.

Evaluations on $A \cup \bigcup_{m < j \leq N} S_j$ span the full dual of $\mathcal{P}_{2N}$. Indeed, if P vanishes there, successive division by the sphere equations gives

$$P = \prod_{j=m+1}^N (q - a_j^2) R, \quad \deg R \leq 2m.$$

To justify each division, write $\xi = (\xi', \xi_n)$ and divide in ξ_n by $|\xi|^2 - r^2$. The remainder is $R_0(\xi') + \xi_n R_1(\xi')$. Its values at $\xi_n = \pm \sqrt{r^2 - |\xi'|^2}$, for $|\xi'| < r$, force both coefficient polynomials to vanish on an open subset of $\mathbb{R}^{n-1}$, hence identically. Distinct sphere factors do not vanish on each other, so division can be repeated. The displayed product is nonzero on A , where R must therefore vanish. Interpolation at level m gives $R = 0$.

Extend the independent evaluations on $A \cup C$ to a basis of this finite-dimensional dual using points of the indicated spheres. The resulting points constitute A' . $\square$

Proof of Theorem 9.1. Work first in the normalization of Lemma 9.2. Choose a countable dense set $\{f_1, f_2, \dots\}$ in $L^2(\Omega)$. Construct strictly increasing integers N_s and nested finite sets A_s , starting from $N_0 = 0$, $A_0 = \{0\}$, so that A_s interpolates at level N_s , all new points have sphere index greater than N_{s-1} , and

$$\text{dist}(f_\ell, \text{span}\{e_\lambda : \lambda \in A_s\}) < 2^{-s} \quad (1 \leq \ell \leq s). \quad (9.10)$$

Given the previous stage, Lemma 9.8 supplies a finite set C of frequencies on later spheres which, together with A_{s-1} , already achieves these finitely many approximations. Lemma 9.9 then supplies A_s and N_s . Enlargement of the finite span preserves the approximations.

Set $\Lambda = \bigcup_s A_s$. Condition (9.10) proves completeness. The set is locally finite: the sphere radii tend to infinity, and spheres with indices $N_{s-1} < j \leq N_s$ are used only at stage s , when finitely many points are selected.

For each $\lambda \in \Lambda$, use the first stage s with $\lambda \in A_s$. Interpolation by V_{N_s} gives $F_\lambda \in V_{N_s}$ with

$$F_\lambda(\mu) = \delta_{\lambda\mu} \quad (\mu \in A_s).$$

Every later frequency lies on a sphere of index greater than N_s , where F_λ vanishes identically. Consequently these Kronecker values hold on all of Λ . Lemma 9.3 identifies F_λ with the Fourier transform of an L^2 function g_λ supported in $\overline{B}$. Extending by zero inside Ω gives

$$\langle g_\lambda, e_\mu \rangle_{L^2(\Omega)} = \delta_{\lambda\mu},$$

proving minimality and the asserted support of the biorthogonals.

Finally, if the original domain is $b + A\Omega$ with A invertible, transport the frequencies to $A^{-\top}\Lambda$. The unitary map $f \mapsto |\det A|^{1/2}f(b + A \cdot)$ sends each transported exponential to a nonzero scalar multiple of the original one. This preserves completeness and minimality, carries the supports of the biorthogonals into the John ellipsoid of the original domain, and preserves local finiteness. $\square$

Remark 9.10. The same selected system is complete and minimal on every measurable set between the inscribed open ellipsoid and Ω , up to null sets: completeness passes to restrictions by zero extension, and the biorthogonals supported in the ellipsoid still work. If the ellipsoid is a proper subset of the convex domain, these biorthogonals cannot be complete in $L^2(\Omega)$. Completeness of the dual family is not part of ordinary minimality.

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